

# Programming Assignment 4

by Paz Liegton

Applied Probability and Statistics. Prof. Lucas Finazzi

## Introduction

This report applies four classical hypothesis tests to a dataset of 219 Argentine license plates recorded by hand from around Villa Urquiza. Each entry stores the digit substring, the letter substring, and the plate format (old: ABC 123; new: AB 123 CD). Part A tests whether digits are uniformly distributed via a chi square goodness of fit test. Part B compares mean digit sums between formats using the Welch  $t$ -test. Part C checks whether both formats draw letters from the same distribution. Part D tests whether consecutive letter pairs arise independently. Part E estimates registration years and tests whether the sample is biased toward older or newer plates. All code uses NumPy, SciPy, Pandas, and Matplotlib. References and sources were key.

## 1 Part A: Are the Digits Uniformly Distributed?

Under  $H_0$ , every digit 0–9 appears with equal probability  $p_k = \frac{1}{10}$ , giving expected count  $E_k = n/10$ . Each plate contributes three digit characters as independent observations; we test old and new formats separately. The test statistic  $\chi^2 = \sum_{k=0}^9 (O_k - E_k)^2 / E_k$  follows  $\chi^2(9)$  under  $H_0$ .

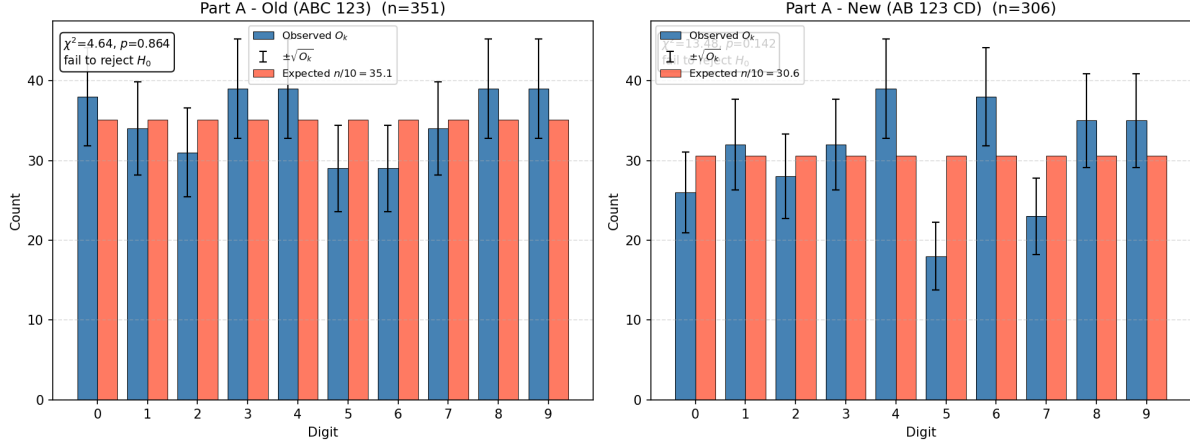


Figure 1: Observed digit counts  $O_k$  (blue) and expected uniform counts  $E_k = n/10$  (red) for old-format plates (left) and new-format plates (right). Error bars are:  $\pm\sqrt{O_k}$ .

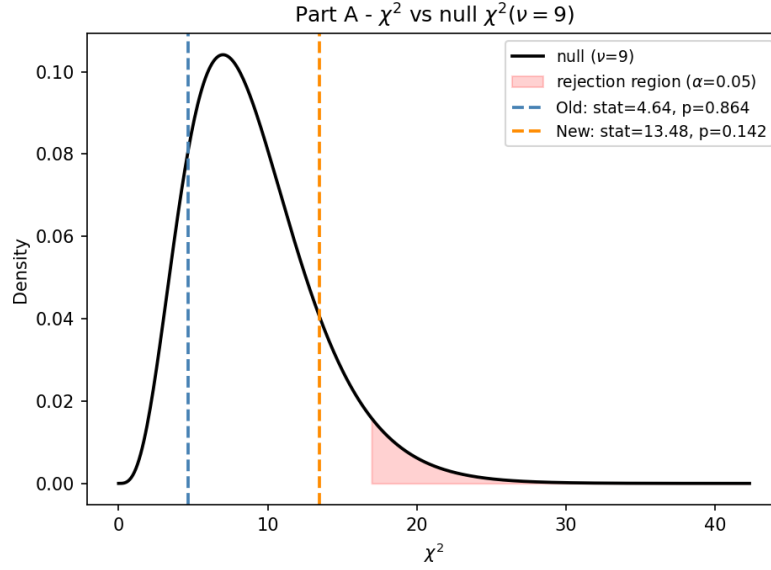


Figure 2: Null  $\chi^2(9)$  density with observed statistics for old (blue) and new (orange) formats. Red shaded region is the rejection tail at  $\alpha = 0.05$ .

At  $\alpha = 0.05$ , both formats fail to reject  $H_0$ . Digit frequencies are consistent with a uniform distribution, suggesting no bias in digit assignment across either era.

## 2 Part B: Do Old and New Format Plates Differ in Mean Digit Sum?

The digit sum  $S = d_1 + d_2 + d_3$  is approximately normal by the CLT ( $\sum$  of three near-uniform variables).  $H_0 : \mu_{\text{old}} = \mu_{\text{new}}$ . Normality is verified via histogram and Q-Q plot before applying Welch's  $t$ -test, which does not assume equal variances before testing.

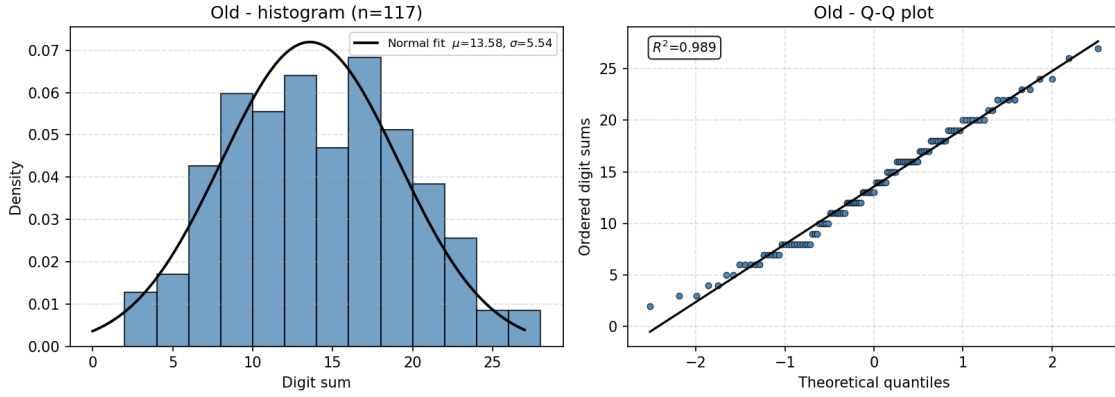


Figure 3: Normality check for old-format digit sums: histogram with Gaussian fit (left) and Q-Q plot (right).

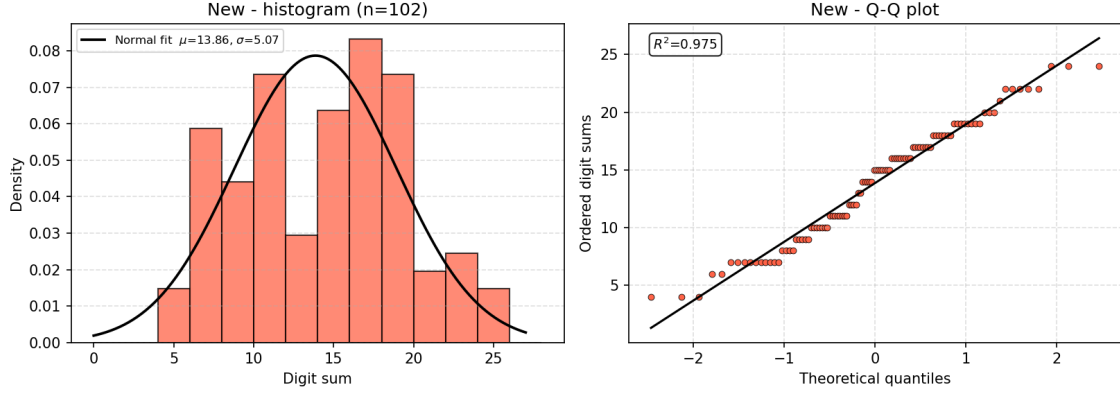


Figure 4: Normality check for new-format digit sums.

The 95% confidence interval for  $\mu_{\text{old}} - \mu_{\text{new}}$  is computed via the Welch–Satterthwaite effective degrees of freedom. Zero lies inside the interval, consistent with the  $t$ -test result: we fail to reject  $H_0$ . There is no significant difference in mean digit sum between the both formats (Figure 5).

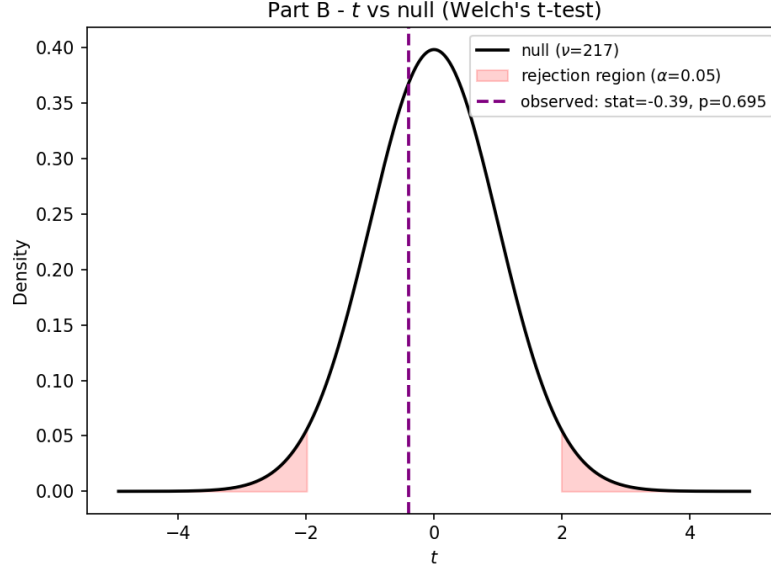


Figure 5: Null  $t(\nu)$  density (two-tailed) with the observed Welch  $t$ -statistic. Red shaded regions are the rejection tails at  $\alpha = 0.05$ .

### 3 Part C: Do Old and New Format Plates Draw Letters from the Same Distribution?

$H_0$ : both formats draw letters from the same distribution. The new-format pool serves as the empirical reference; expected old-format counts are  $E_\ell = n_{\text{old}} \cdot \hat{p}_\ell^{\text{new}}$ . Bins with  $E_\ell < 5$  are pooled into an “other” bin; letters absent from both pools (e.g. I, O, Q, excluded by Argentine traffic law) are dropped. This latter was a review done post oral as there was no research taken about characters being confused for numbers.

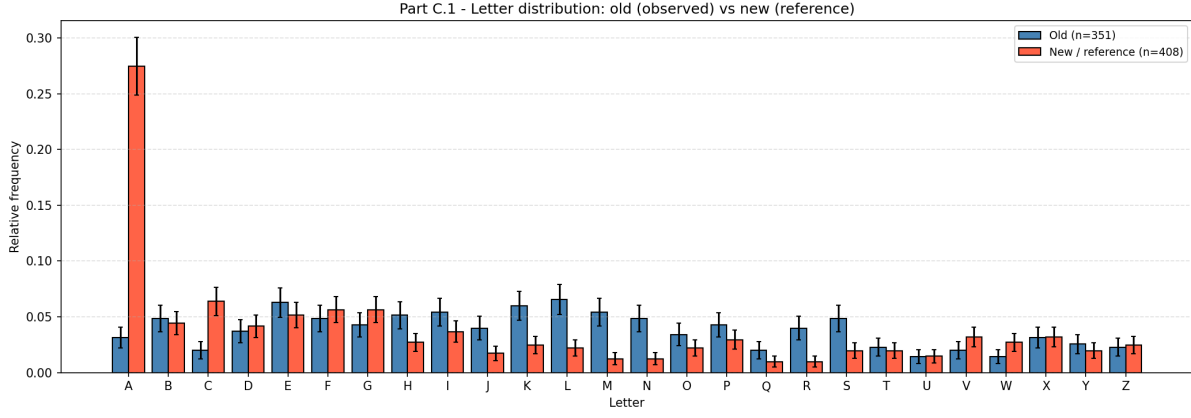


Figure 6: Relative letter frequencies for old-format (blue) and new-format (red) plates, with Poisson error bars  $\pm\sqrt{O_\ell}/n$ .

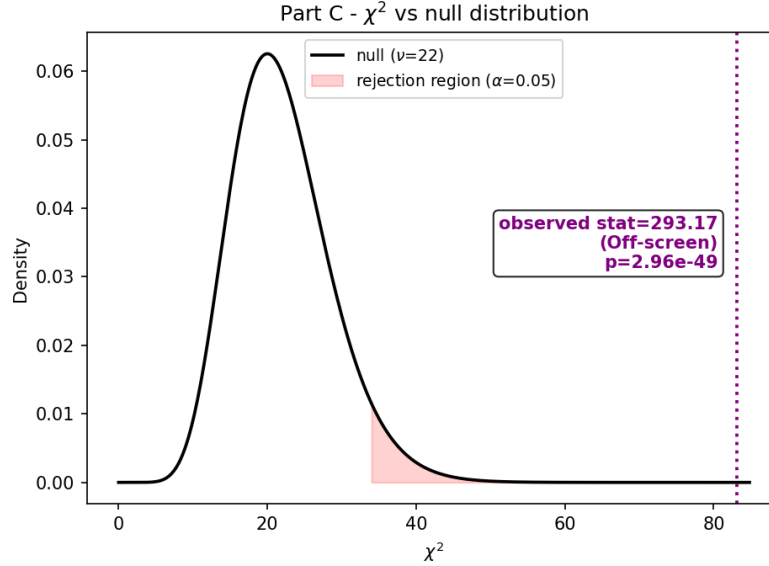


Figure 7: Null  $\chi^2(\nu)$  density with the observed statistic from Part C.

$H_0$  is rejected. Figure 8 (as shown later) shows signed standardised residuals  $(O_\ell - E_\ell)/\sqrt{E_\ell}$  per letter, with the top-five contributors highlighted in red. The dashed lines mark  $\pm 2$  (per-bin 95% Poisson bands for a single test).

With  $K = 26$  bins, however, the Bonferroni-corrected threshold is  $\pm z_{\alpha/2K} = \pm z_{0.05/52} \approx \pm 3.29$ : individual letters are only flagged as significant after multiple-comparison correction if  $|r| > 3.29$ . Several letters (A, M, N, R, L) comfortably exceed this threshold, confirming that the rejection of  $H_0$  is not an artefact of multiple comparisons. The distributional shift reflects the alphabetic issuance order: old and new formats cover distinct ranges of the alphabet. This is of course due to issues being done in a yearly basis, and old characters running out.

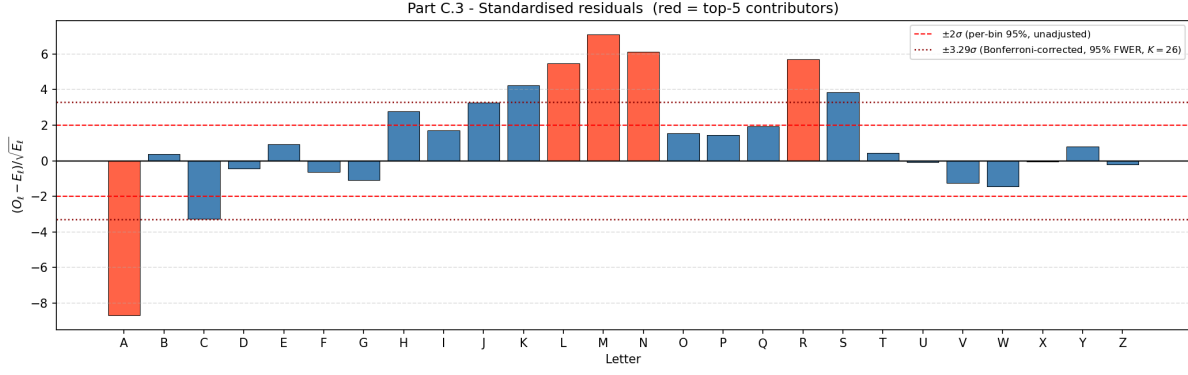


Figure 8: Standardised residuals per letter. Dashed red lines mark the unadjusted  $\pm 2\sigma$  per-bin band; dotted dark red lines mark the Bonferroni-corrected  $\pm 3.29\sigma$  threshold (95% FWER,  $K = 26$ ). Red bars are the five largest contributors to  $\chi^2$ . Note that this is something new I tried to do, references are on the bottom.

## 4 Part D: Are Certain Letter Pairs Suppressed?

$H_0$ : consecutive letters are drawn independently, so  $P(j, k) = \hat{p}_j \cdot \hat{p}_k$  and  $E_{jk} = N\hat{p}_j\hat{p}_k$ . Each old plate contributes 2 pairs (AB, BC); each new plate contributes 3 (AB, BC, CD). The test is restricted to  $K$  letters with  $\geq 10$  individual occurrences, ensuring  $E_{jk} \geq 5$  throughout;  $\nu = K^2 - 1$ .

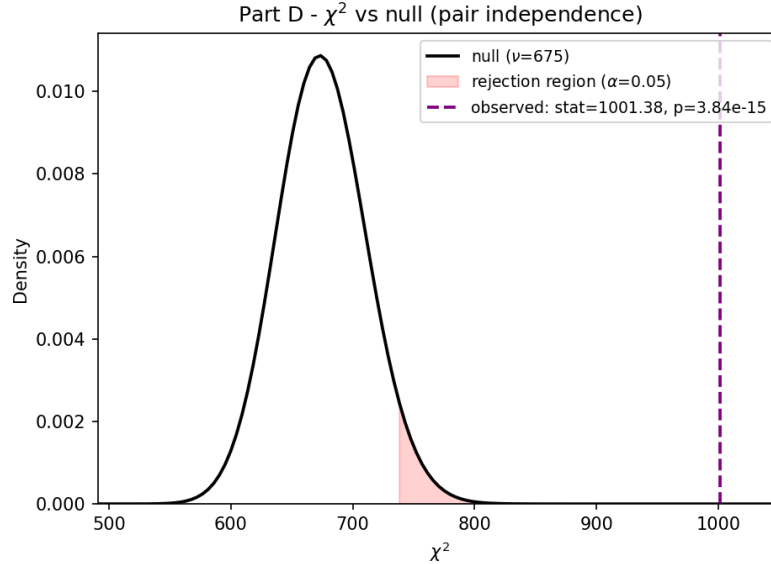


Figure 9: Null  $\chi^2(K^2 - 1)$  density rescaled to  $\pm 5\sigma$  around the null mean due to a suggestion by the professor, with the observed statistic from Part D marked. The null is centred at  $\nu \approx 675$ ; the observed statistic at  $\approx 1001$  falls far into the rejection region ( $p = 3.84 \times 10^{-15}$ ).

$H_0$  is strongly rejected. Figure 10 shows the full  $K \times K$  standardised residual matrix; the five most over- and under-represented pairs are outlined in black. Under-represented pairs are not as apparent for profane words, maybe the dataset is not big enough yet.

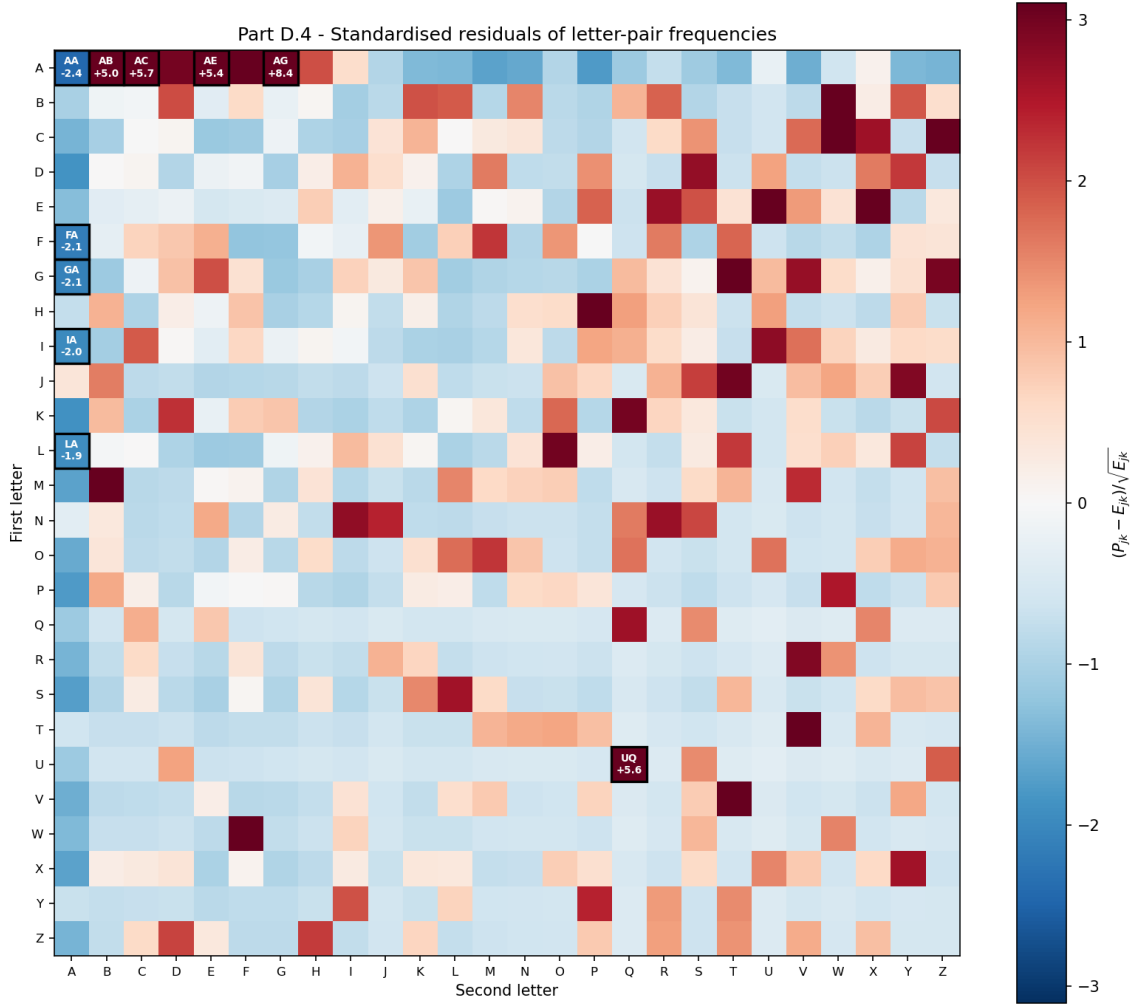


Figure 10: Standardised residual matrix  $(P_{jk} - E_{jk})/\sqrt{E_{jk}}$ . Blue = under-represented; red = over-represented. Black outlines mark the five most extreme pairs in each direction.

## 5 Part E: Is Our Sample Biased Toward Older or Newer Plates?

Each plate's registration year is estimated by mapping its alphanumeric code to a known issuance milestone via linear interpolation, using anchor data from Wikipedia's Argentine license plate article.

### My Null Hypothesis

$H_0$ : the mean estimated year equals the midpoint of the full Argentine issuance range. If our 219 plates were a uniform random draw from every plate ever issued, the expected mean would be

$$\mu_0 = \frac{\text{earliest anchor} + \text{collection year}}{2} = \frac{1995 + 2026}{2} = 2010.5.$$

A significant result indicates that the street sample is temporally skewed:  $t > 0$  means over-representation of newer plates;  $t < 0$  means older plates dominate.

## My Results

We apply a one-sample  $t$ -test with  $\nu = n - 1$  degrees of freedom and report the 95% CI on the mean.

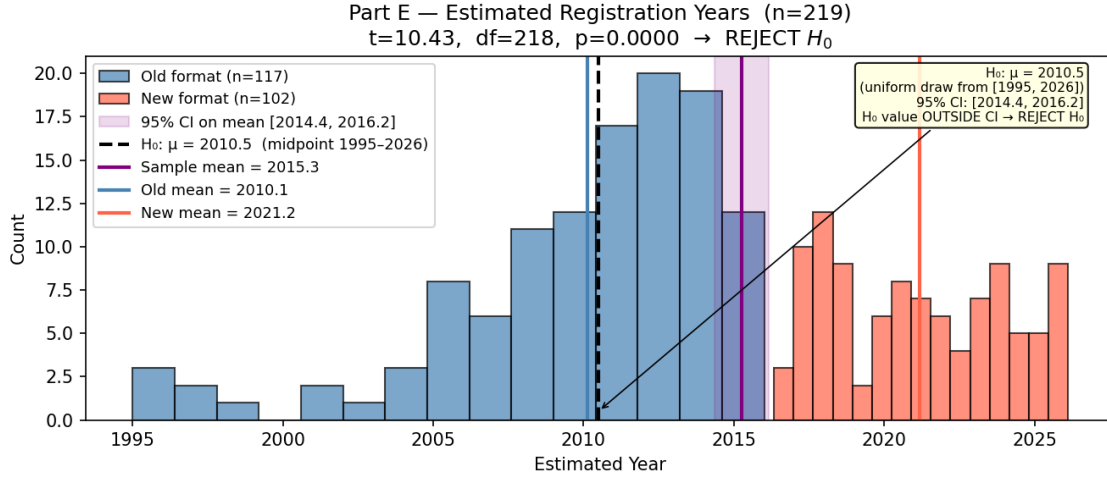


Figure 11: Estimated registration years for old-format (blue) and new-format (red) plates. The dashed black line is  $H_0 : \mu_0 = 2010.5$  (midpoint 1995-2026); the purple shaded band is the 95% CI on the sample mean. If the dashed line falls outside the band,  $H_0$  is rejected. Annotations boxes were a nightmare.

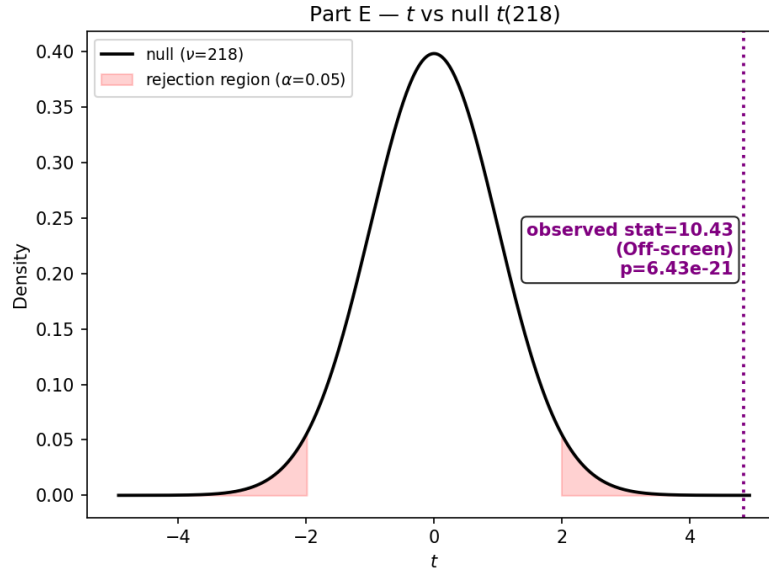


Figure 12: Null  $t(\nu)$  density (two-tailed) with the observed  $t$ -statistic for Part E.

If  $H_0$  is rejected and  $t > 0$ , the sample skews toward newer vehicles an expected result for a street convenience sample, where recently registered cars are more common.

## Conclusion

Digit frequencies are consistent with a uniform distribution and mean digit sums do not differ between formats (Parts A–B), suggesting no detectable administrative pattern in digit assignment. Letter distributions differ significantly between formats (Part C), reflecting their distinct alphabetic issuance ranges;

several letters remain significant even after Bonferroni correction. Consecutive letter pairs deviate strongly from independence (Part D), sort of consistent with a deliberate suppression policy for certain syllable combinations, more data is needed. Part E tests whether the street sample is temporally representative of the full plate registry. (See Slides I provided to see the comparison with Argentinian vehicles)

## Biographies/Sources.

- Cowan, Glen. *Statistical Data Analysis in Particle Physics*. Oxford University Press, 1998.
- SciPy Documentation. `scipy.stats` module. <https://docs.scipy.org>
- Wikipedia. “Vehicle registration plates of Argentina.” [https://en.wikipedia.org/wiki/Vehicle\\_registration\\_plates\\_of\\_Argentina](https://en.wikipedia.org/wiki/Vehicle_registration_plates_of_Argentina)
- Wikipedia. “Bonferroni correction.” [https://en.wikipedia.org/wiki/Bonferroni\\_correction](https://en.wikipedia.org/wiki/Bonferroni_correction)
- AI Assistance: Claude (Anthropic). Helped with formatting, L<sup>A</sup>T<sub>E</sub>X structuring, and statistical methodology documentation.
- Python Graph Gallery. Matplotlib examples. <https://python-graph-gallery.com>